

Dhananjay Thosar

Data Engineer | Python | PySpark | SQL | AWS | ETL/ELT | Data Warehousing

Hyderabad, India | +91 8390833406 | thosardhananjay25@gmail.com | linkedin.com/in/dhananjay-thosar-38b5641b9

PROFESSIONAL SUMMARY

Data Engineer with **3+ years of experience** designing, building and supporting cloud-based data pipelines, integrations and data platforms on **AWS**. Skilled in Python, PySpark, SQL and ETL/ELT development, dimensional modelling (star and snowflake schema) and data warehousing on Amazon Redshift. Experienced embedding data quality, governance and security into production workloads, providing operational support and troubleshooting, and partnering with business and technology stakeholders to deliver reliable analytics-ready datasets. AWS and Databricks certified; **4x Berkadia Award** winner.

TECHNICAL SKILLS

Programming: Python, SQL, PySpark, Pandas

AWS: S3, Glue, Lambda, Amazon Redshift, EMR, Athena, Step Functions, CodePipeline, CloudWatch, IAM, SNS, EventBridge

Data Engineering: ETL/ELT, data pipelines, data integration, CDC, batch processing, metadata-driven ingestion, data reconciliation

Data Warehousing & Modelling: Amazon Redshift, dimensional modelling, star schema, snowflake schema, SCD Type 1 & 2, AWS Glue Data Catalog, Apache Parquet

Orchestration & DevOps: AWS Step Functions, Git/GitHub, CI/CD, Agile

Analytics & Reporting Support: Amazon Athena, Power BI

Generative AI: LLM, RAG, Prompt Engineering, embeddings/vector stores (FAISS), LangChain fundamentals

PROFESSIONAL EXPERIENCE

Associate Data Engineer

Berkadia India Pvt. Ltd.

Jul 2023 – Present

Hyderabad, India

Platinum Enterprise Data Platform — AWS Data Pipelines *PySpark · AWS EMR · Amazon S3 · AWS Glue · Amazon Redshift · Amazon Athena · AWS Lambda · AWS Step Functions*

- Built and productionized PySpark and Spark SQL pipelines on AWS EMR, delivering curated Property, Account, Contact, Loan and Sales data domains from **14+ upstream source systems** into Amazon S3 and Amazon Redshift for enterprise reporting and analytics.
- Designed a layered ETL/ELT architecture on Amazon S3, standardizing heterogeneous raw feeds into validated, de-duplicated curated datasets aligned with enterprise data standards.
- Implemented incremental processing using Apache Parquet on Amazon S3, AWS Glue jobs and MERGE-based upserts, replacing full-refresh loads and improving pipeline reliability and runtime efficiency.
- Built an event-driven, metadata-driven ingestion framework using AWS Lambda and AWS Step Functions (validation, threshold checks, retries, Amazon SNS alerting) so new sources onboard via configuration rather than custom code.
- Modelled analytics datasets using dimensional modelling (star schema, SCD Type 2) and published curated tables through AWS Glue Data Catalog, Amazon Athena and Amazon Redshift to support Power BI dashboards and business reporting.
- Tuned PySpark workloads on AWS EMR (partitioning, broadcast joins, file sizing) and embedded automated data quality checks, improving job performance and lowering compute cost.

MDM Modernization, Cloud Migration & Data Reconciliation

PySpark · Apache Spark · SQL · AWS EMR · AWS Glue · Python

- Migrated **20+ legacy SQL stored procedures** into modular PySpark applications on AWS EMR as part of a Master Data Management (MDM) modernization programme.
- Re-engineered transformation, matching and aggregation logic to process **500+ GB of data daily** through distributed PySpark processing, replacing sequential row-based SQL execution.
- Reduced end-to-end processing time by roughly **3 hours** by redesigning workloads for parallel execution and optimizing joins and aggregations, using Spark UI benchmarking to guide performance tuning.
- Built a reusable Python/PySpark/SQL data reconciliation framework validating schema, datatype and record-level parity between legacy and modernized platforms, and owned migration verification through legacy decommissioning. Recognised with a **Berkadia Leaf Award**.

EMR Cluster Optimization & Ad-hoc Job Automation Framework

AWS EMR · AWS Step Functions · AWS Lambda · Amazon SNS ·

Python

- Replaced a fully manual AWS EMR workflow (infra approvals, PuTTY access, manual spark-submit) costing roughly **1 hour of overhead per ad-hoc run** with an automated AWS Step Functions framework.
- Built a Lambda-based execution interface letting engineers trigger parameterised PySpark jobs directly from the AWS Console, cutting provisioning-to-termination time from **~1 hour to a few minutes**.
- Adopted as a self-service utility across the team; recognised with a **Berkadia HiFi Award** for operational efficiency and continuous improvement impact.

Software Developer Intern – Full Stack Development

Nov 2021 – May 2022

Technical Support and Software Committee, PICT

Pune, India

- Owned end-to-end database design and development (PHP, MySQL) for a hostel management system handling student records, warden data, room allocation, and fee processing — gaining real, hands-on SQL experience across the full data lifecycle.
- Engineered a region-based room allocation module, allowing students to filter and view available rooms by hometown/region (e.g., Amravati) to prioritize roommate preferences, using structured queries across student and room-availability tables.
- Implemented secure OTP-based verification and automated fee receipt generation, integrating transactional logic with the database to ensure accurate, real-time record updates.
- Designed relational schemas with foreign key constraints and cascading deletes (ON DELETE CASCADE) — e.g., auto-removing library and canteen records on admission cancellation — ensuring referential integrity across interconnected modules.

AI / ML Intern

Aug 2021 – Sep 2021

Elite Techno Groups

Remote

- Built Python-based machine learning solutions spanning data preprocessing, feature engineering, model training and evaluation.

PERSONAL PROJECTS

SchemaMind — RAG Copilot for Data Docs

Jul 2026 – Aug 2026

Python · LangChain · LLM · FAISS · FastAPI · Ollama

- Built a retrieval-augmented generation (RAG) assistant that answers natural-language questions over pipeline documentation, data dictionaries and table schemas.
- Implemented document chunking, embedding generation and semantic search over a FAISS vector store, with prompt templates grounding LLM responses in retrieved context to avoid hallucinated schema details.
- Delivered a local FastAPI chat UI and CLI.
- GitHub: github.com/Dhananjaythosar/schemamind

EDUCATION

Bachelor of Engineering, Computer Engineering

2019 – 2023

Pune Institute of Computer Technology (PICT), Pune, India

CGPA: 9.2 / 10

CERTIFICATIONS

- **AWS Certified Data Engineer — Associate** [View credential](#)
- **Databricks Certified Associate Developer for Apache Spark 3.0** [View credential](#)
- **AWS Certified Cloud Practitioner** [View credential](#)

ACHIEVEMENTS

- **Berkadia HiFi Award — 3 Times** for delivery excellence and impactful data engineering initiatives.
- **Berkadia Leaf Award** for the MDM modernization and legacy decommissioning programme.